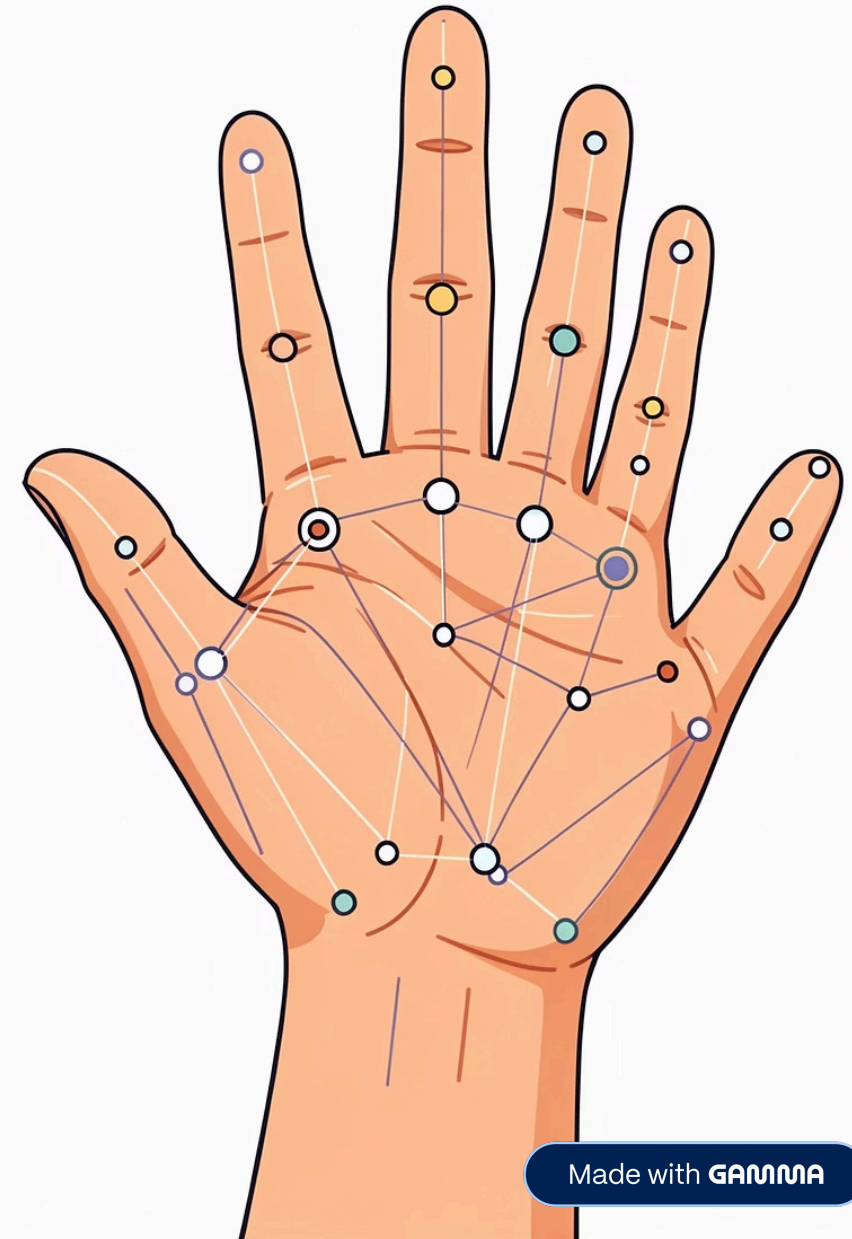


Gesture-Controlled System using Computer Vision and Machine Learning

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Problem Statement

- Traditional interaction paradigms require physical input devices such as mice, keyboards, and touchscreens
- Growing demand for touchless, intuitive interfaces—particularly in healthcare, public spaces, and accessibility contexts
- Primary goal: enabling reliable application control through natural hand gestures without physical contact
- System architecture employs real-time webcam-based acquisition and processing for practical deployment scenarios

System Architecture Overview

- Webcam captures real-time video stream (~30 FPS)
- MediaPipe framework detects 21 hand landmarks in real time
- Feature extraction pipeline converts spatial coordinates into 63-dimensional vectors
- Two parallel classification approaches evaluated: Rule-Based threshold system and k-Nearest Neighbors classifier
- Application control signals generated through normalized gesture-to-command mapping

Feature Engineering Methodology

- 21 hand landmarks tracked per frame: wrist, MCP joints, PIP joints, DIP joints, and fingertips
- Each landmark represented by (x, y, z) Cartesian coordinates
- Wrist-centered transformation eliminates translational variance
- Scale normalization applied using wrist-to-middle MCP distance
- Final feature vector dimensionality: 63 floating-point values (21 landmarks $\times$ 3 coordinates)
- Rotation robustness improved through distance-based and scale-invariant feature normalization

Rule-Based Classification Approach

- Finger extension detection implemented through Euclidean distance calculations between fingertip and MCP joint pairs
- Geometric threshold rules defined for each gesture class based on anatomical hand configurations
- No statistical training required; only manual threshold tuning
- **Advantages:** Interpretable decision logic, minimal computational overhead, immediate deployment
- **Limitations:** Sensitive to threshold selection, poor generalization across users and scale variations, requires manual rule refinement

Machine Learning Approach

- **Algorithm:** k-Nearest Neighbors with $k = 7$ (odd value prevents tie votes)
- **Dataset:** 304 manually collected gesture samples with manually labeled gesture samples
- **Dataset distribution:** FIST (60), OPEN_PALM (61), POINT_LEFT (61), POINT_RIGHT (61), THUMBS_UP (61)
- **Data collection protocol:** Single user, controlled lighting, consistent camera positioning, Approximately balanced class distribution (~60 samples per class)
- **Validation:** 80/20 train-test split with stratification ensuring class representation
- **Implementation:** Feature normalization applied during preprocessing

Offline Evaluation Results

- **Test set accuracy:** 86.67% on 60 unseen samples (52/60)
- **Train/Test split:** 80/20 (304 total samples)
- Confusion matrix analyzed per gesture class
- Temporal smoothing not applied during offline evaluation
- Model generalizes to unseen test samples

Live Real-Time Evaluation

- **Trial protocol:** 100 trials conducted per method under identical conditions
- **Environmental controls:** Same user, consistent lighting, fixed camera position
- **Stability filtering** implemented (5-frame consistency check)
- **Rule-based performance:** 75% accuracy (75/100 correct)
- **k-NN performance:** 95% accuracy (95/100 correct)
- **Real-time performance** achieved on standard consumer hardware

Live Real-Time Evaluation

LIVE EVALUATION

Mode: RULE (press m)

PRED: NO_HAND

Acc: 75.0% (75/100) r=reset ESC=quit

LIVE EVALUATION

Mode: ML (press m)

PRED: NO_HAND

Acc: 95.0% (95/100) r=reset ESC=quit

Performance Comparison

Metric	Rule-Based	Machine Learning (k-NN)
Training Required	No	Yes (304 samples, 80/20 split)
Offline Accuracy	N/A	86–87%
Live Accuracy (100 trials)	75%	95%
Robustness to Variation	Low (threshold-sensitive)	Moderate–High (data-driven, validated on test split)
Generalization Ability	Limited	Demonstrated via unseen test split (86.7%)
Implementation Complexity	Low (geometric rules)	Medium (pipeline orchestration)

Conclusion & Future Work

- **Technical achievement:** ML approach achieved 20 percentage point improvement in live accuracy (95% vs 75%)
- **Robustness validation:** Experimental results indicate improved robustness to user variation compared to fixed-threshold rules
- **Limitation analysis:** Rule-based systems constrained by fixed geometric thresholds unable to accommodate individual hand morphology differences
- **Future directions:**
 - Expanded dataset collection across multiple users with diverse hand sizes and morphologies
 - Temporal smoothing and sequence modeling using RNN architectures
 - Exploration of deep learning architectures (e.g., CNNs) using raw image inputs to eliminate manual feature engineering
 - Cross-user validation through leave-one-user-out testing protocols
 - Real-world deployment testing with variable lighting and camera hardware